

## py.Aroma: An Intuitive Graphical User Interface for Diverse Aromaticity Analyses

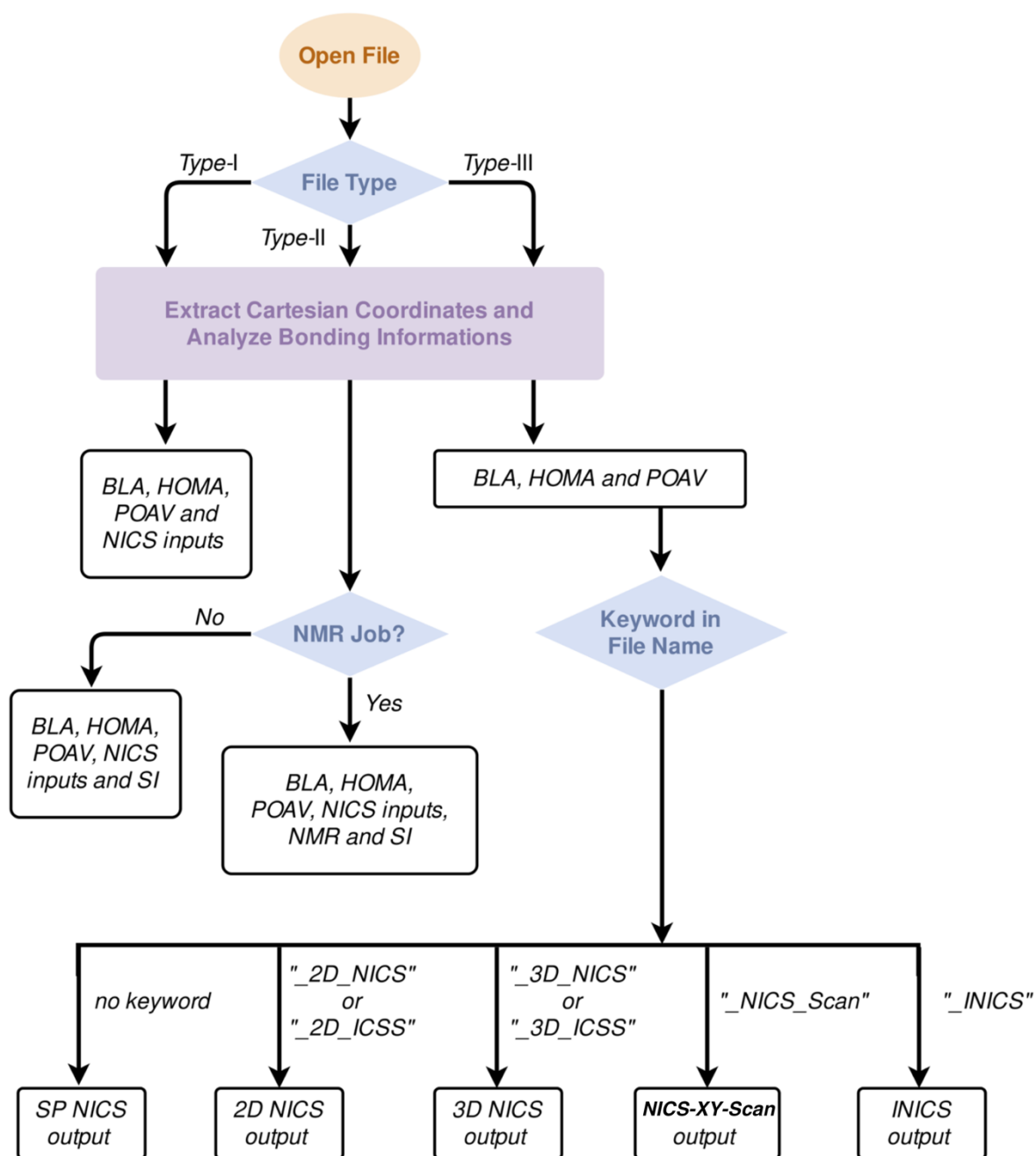
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## 1 Additional Information about py.Aroma



**Figure S1.** Workflow of py.Aroma.

Gaussian Calculation Setup

Job Type

NMR Method: GIAO Method

☐ Compute spin-spin couplings (for all atoms)

☐ NMR=mixed

Calculation Method

Method: Default Spin B3LYP

Basis Set: 6-311G + ( 2d , p )

Charge: 0

Spin: Singlet

Link 0

☒ Memory Limit: 10 GB ☒ Processors: 8

☒ Chkpoint File Default Name

☐ Other:

General

☐ Use Quadratically Convergent SCF ☐ Ignore Symmetry ☒ Additional Print

Guess

Guess Method: Default

☐ Mix HOMO and LUMO orbitals ☐ Only do guess (no SCF) ☐ Localize orbitals ☐ Save orbitals to chk file

Solvation

Model: None

Solvent: Default

Additional Keywords

Default Preview Save Close

```

%nprocshared=8
%mem=10gb
%chk=filename.chk
#p nmr=giao b3lyp/6-311+g(2d,p)

Title

0 1
  
```

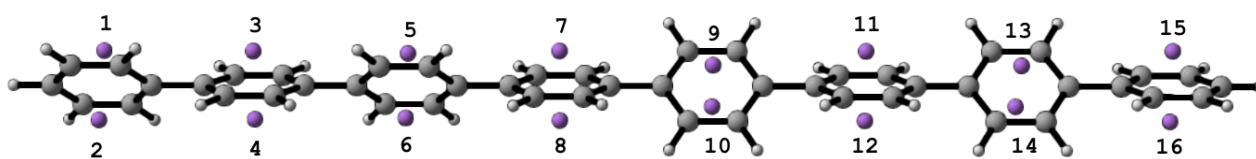
**Figure S2.** Screenshot of “Gaussian Calculation Setup” window in py.Aroma.

## 2 Examination of Validity of NICS<sub>⊥</sub> for [8]CPP and [8]LPP

To verify the validity of the NICS(1)<sub>⊥</sub> metric, the NICS(1) and NICS(−1) values for [8]LPP and [8]CPP are summarized in Tables S1 and S2, and the geometries used for calculations are shown in Figure S3 and S4, respectively. These values were calculated at the B3LYP/6-31+G(d) level of theory.

For [8]LPP, the NICS(±1)<sub>⊥</sub> values of approximately −25 ppm for the phenyl rings indicate their aromaticity, comparable to that of pristine benzene (−28.75 ppm). For phenyl rings parallel to the XY plane, NICS(±1)<sub>⊥</sub> values closely reproduce the NICS(±1)<sub>ZZ</sub> values.

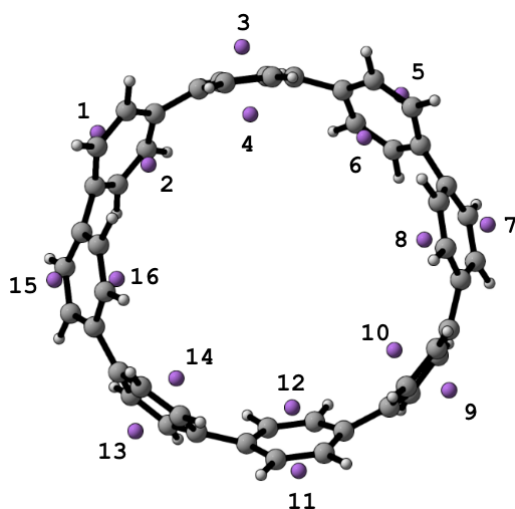
For [8]CPP, the NICS(1)<sub>⊥</sub> values on the outer side of the CPP ring are approximately −25 ppm, reflecting the aromaticity of the phenyl rings. On the other hand, the NICS(−1)<sub>⊥</sub> values on the inner side of the CPP ring are around −22 ppm, indicating slightly weaker aromaticity.



**Figure S3.** Geometry of [8]LPP with Bq atoms used for NICS calculations. Note that the 1st, 3rd, 5th, and 7th phenyl rings (from the left) are parallel to the XY plane.

**Table S1.** NICS(±1) values for all tensor components for each phenyl ring in [8]LPP.

| Bq No. | NICS Tensor Component / ppm |       |       |       |        |        |       |        |        |        |
|--------|-----------------------------|-------|-------|-------|--------|--------|-------|--------|--------|--------|
|        | XX                          | YX    | ZX    | XY    | YY     | ZY     | XZ    | YZ     | ZZ     | ⊥      |
| 1      | −1.61                       | 0.12  | 0.45  | 0.64  | −0.41  | 0.53   | 0.16  | 0.18   | −26.49 | −26.49 |
| 2      | −1.61                       | −0.12 | −0.45 | −0.64 | −0.41  | 0.53   | −0.16 | 0.18   | −26.49 | −26.49 |
| 3      | −2.36                       | 0.04  | 0.03  | 0.02  | −11.54 | −12.10 | 0.22  | −12.11 | −12.96 | −24.32 |
| 4      | −2.36                       | −0.04 | −0.03 | −0.02 | −11.54 | −12.10 | −0.22 | −12.11 | −12.96 | −24.32 |
| 5      | −2.51                       | −0.01 | 0.01  | −0.04 | 0.06   | 0.89   | −0.28 | 0.87   | −24.01 | −23.99 |
| 6      | −2.51                       | 0.01  | −0.01 | 0.04  | 0.06   | 0.89   | 0.28  | 0.87   | −24.01 | −23.99 |
| 7      | −2.51                       | −0.01 | 0.00  | −0.08 | −11.27 | −12.19 | −0.03 | −12.15 | −12.79 | −24.17 |
| 8      | −2.51                       | 0.01  | 0.00  | 0.08  | −11.27 | −12.19 | 0.03  | −12.15 | −12.79 | −24.17 |
| 9      | −2.45                       | −0.21 | 0.00  | −1.46 | 0.24   | −0.04  | 0.07  | −0.13  | −24.06 | −24.06 |
| 10     | −2.45                       | 0.21  | 0.00  | 1.46  | 0.24   | −0.04  | −0.07 | −0.13  | −24.06 | −24.06 |
| 11     | −2.46                       | 0.01  | 0.00  | −0.08 | −11.20 | 12.22  | −0.06 | 12.17  | −12.95 | −24.25 |
| 12     | −2.46                       | −0.01 | 0.00  | 0.08  | −11.20 | 12.22  | 0.06  | 12.17  | −12.95 | −24.25 |
| 13     | −2.46                       | −0.01 | −0.05 | −0.10 | −0.31  | −0.98  | 0.38  | −0.90  | −24.22 | −24.20 |
| 14     | −2.46                       | 0.01  | 0.05  | 0.10  | −0.31  | −0.98  | −0.38 | −0.90  | −24.22 | −24.20 |
| 15     | −1.22                       | 0.21  | −0.42 | −0.21 | −11.84 | 13.23  | −1.04 | 13.04  | −14.36 | −26.29 |
| 16     | −1.22                       | −0.21 | 0.42  | 0.21  | −11.84 | 13.23  | 1.04  | 13.04  | −14.36 | −26.29 |



**Figure S4.** Geometry of [8]CPP with Bq atoms used for NICS calculations.

**Table S2.** NICS( $\pm 1$ ) values for all tensor components for each phenyl ring in [8]CPP.

| Bq No. | NICS Tensor Component / ppm |        |        |        |        |        |        |        |       |         |
|--------|-----------------------------|--------|--------|--------|--------|--------|--------|--------|-------|---------|
|        | XX                          | YY     | ZZ     | XY     | YX     | ZX     | XZ     | YZ     | ZY    | $\perp$ |
| 1      | -9.44                       | -9.80  | 5.28   | -9.69  | -8.81  | 4.54   | 5.63   | 5.12   | -5.24 | -22.00  |
| 2      | -15.30                      | -8.15  | 6.66   | -7.81  | -14.40 | 7.06   | 6.99   | 7.99   | -1.85 | -26.93  |
| 3      | -16.90                      | -0.35  | -7.36  | -0.81  | 0.67   | -0.79  | -7.84  | -0.74  | -6.52 | -20.92  |
| 4      | -20.16                      | -0.29  | -10.44 | -0.85  | -6.87  | -0.12  | -11.41 | 0.69   | -3.90 | -25.66  |
| 5      | -9.35                       | 10.71  | 0.43   | 11.00  | -10.33 | 0.87   | 0.40   | 0.65   | -2.89 | -20.36  |
| 6      | -15.68                      | 8.73   | -1.01  | 9.22   | -16.40 | 0.59   | -2.07  | -1.34  | 1.95  | -24.63  |
| 7      | 0.42                        | 0.00   | 0.00   | 0.00   | -15.10 | -8.17  | 0.00   | -8.71  | -7.24 | -20.47  |
| 8      | -6.60                       | 0.00   | 0.00   | 0.00   | -17.97 | -11.19 | 0.00   | -12.38 | -5.51 | -25.07  |
| 9      | -9.34                       | -10.71 | -0.43  | -11.00 | -10.34 | 0.87   | -0.40  | 0.65   | -2.89 | -20.36  |
| 10     | -15.67                      | -8.73  | 1.01   | -9.22  | -16.41 | 0.59   | 2.07   | -1.33  | 1.95  | -24.63  |
| 11     | -16.90                      | 0.34   | 7.36   | 0.80   | 0.68   | -0.79  | 7.83   | -0.74  | -6.52 | -20.92  |
| 12     | -20.17                      | 0.29   | 10.44  | 0.84   | -6.87  | -0.11  | 11.40  | 0.70   | -3.89 | -25.65  |
| 13     | -9.45                       | 9.79   | -5.28  | 9.69   | -8.81  | 4.54   | -5.63  | 5.12   | -5.25 | -22.00  |
| 14     | -15.30                      | 8.15   | -6.66  | 7.80   | -14.39 | 7.06   | -7.00  | 7.99   | -1.85 | -26.93  |
| 15     | 0.43                        | 0.00   | 0.00   | 0.00   | -19.93 | -5.27  | 0.00   | -6.12  | -3.53 | -21.69  |
| 16     | -7.05                       | 0.00   | 0.00   | 0.00   | -24.21 | -7.11  | 0.00   | -7.90  | 0.22  | -26.33  |

### 3 Cartesian Coordinates of Bq Atoms Generated by NICS Modules using py.Aroma

This section provides examples from Figures 3a, 3b, and 3e of the original manuscript, showing the Cartesian coordinates of Bq atoms used for NICS calculations. These coordinates were generated by the NICS module in py.Aroma.

#### 3.1 NICS(0) and NICS( $\pm 1$ ) for Benzene

|    |           |           |           |
|----|-----------|-----------|-----------|
| C  | -1.339000 | -0.396000 | 0.000000  |
| C  | -0.327000 | -1.358000 | 0.000000  |
| C  | 1.012000  | -0.962000 | 0.000000  |
| C  | 1.339000  | 0.396000  | 0.000000  |
| C  | 0.327000  | 1.358000  | 0.000000  |
| C  | -1.013000 | 0.962000  | 0.000000  |
| H  | -2.381000 | -0.704000 | 0.000000  |
| H  | -0.581000 | -2.414000 | 0.000000  |
| H  | 1.800000  | -1.710000 | 0.000000  |
| H  | 2.381000  | 0.704000  | 0.000000  |
| H  | 0.581000  | 2.414000  | 0.000000  |
| H  | -1.800000 | 1.710000  | 0.000000  |
| Bq | -0.000167 | 0.000000  | 1.000000  |
| Bq | -0.000167 | 0.000000  | -1.000000 |
| Bq | -0.000167 | 0.000000  | 0.000000  |

#### 3.2 NICS(1)-XY-Scan for Pentalene

|    |           |           |           |
|----|-----------|-----------|-----------|
| H  | -1.524521 | 2.260771  | 0.000524  |
| C  | -1.214429 | 1.221474  | 0.000304  |
| C  | 0.039303  | 0.729078  | -0.000128 |
| C  | -0.039303 | -0.729080 | -0.000142 |
| C  | -1.468332 | -1.092067 | 0.000166  |
| C  | -2.162834 | 0.064262  | -0.000301 |
| H  | -1.876015 | -2.094454 | 0.000238  |
| H  | -3.241330 | 0.167021  | -0.000485 |
| C  | 1.214430  | -1.221473 | 0.000147  |
| C  | 2.162834  | -0.064262 | -0.000137 |
| C  | 1.468331  | 1.092068  | 0.000004  |
| H  | 1.524523  | -2.260771 | 0.000355  |
| H  | 3.241330  | -0.167021 | -0.000153 |
| H  | 1.876011  | 2.094455  | 0.000041  |
| Bq | -2.162834 | 0.064262  | 1.000000  |

|    |           |           |          |
|----|-----------|-----------|----------|
| Bq | -1.962922 | 0.058322  | 1.000000 |
| Bq | -1.763010 | 0.052382  | 1.000000 |
| Bq | -1.563099 | 0.046443  | 1.000000 |
| Bq | -1.363187 | 0.040503  | 1.000000 |
| Bq | -1.163275 | 0.034563  | 1.000000 |
| Bq | -0.963363 | 0.028623  | 1.000000 |
| Bq | -0.763452 | 0.022684  | 1.000000 |
| Bq | -0.563540 | 0.016744  | 1.000000 |
| Bq | -0.363628 | 0.010804  | 1.000000 |
| Bq | -0.163716 | 0.004864  | 1.000000 |
| Bq | 0.036196  | -0.001075 | 1.000000 |
| Bq | 0.236107  | -0.007015 | 1.000000 |
| Bq | 0.436019  | -0.012955 | 1.000000 |
| Bq | 0.635931  | -0.018895 | 1.000000 |
| Bq | 0.835843  | -0.024835 | 1.000000 |
| Bq | 1.035754  | -0.030774 | 1.000000 |
| Bq | 1.235666  | -0.036714 | 1.000000 |
| Bq | 1.435578  | -0.042654 | 1.000000 |
| Bq | 1.635490  | -0.048594 | 1.000000 |
| Bq | 1.835402  | -0.054533 | 1.000000 |
| Bq | 2.035313  | -0.060473 | 1.000000 |

### 3.3 INICS for 1-Bromoethylbenzene

|   |           |           |           |
|---|-----------|-----------|-----------|
| H | 1.362407  | 1.739433  | 1.549908  |
| C | 1.653354  | 0.937937  | 0.875138  |
| C | 2.403409  | -1.114809 | -0.839587 |
| C | 0.701971  | 0.365681  | 0.032077  |
| C | 2.973204  | 0.492484  | 0.860092  |
| C | 3.351120  | -0.533699 | 0.001786  |
| C | 1.088571  | -0.668565 | -0.824343 |
| H | 3.702928  | 0.946507  | 1.523812  |
| H | 4.378365  | -0.885415 | -0.009424 |
| H | 0.345380  | -1.139501 | -1.462118 |
| H | 2.690269  | -1.922764 | -1.506014 |
| C | -0.697958 | 0.922224  | 0.006207  |
| H | -0.848261 | 1.591830  | 0.853998  |
| C | -1.039059 | 1.632509  | -1.296874 |
| H | -0.945840 | 0.953610  | -2.149201 |

|    |           |           |           |
|----|-----------|-----------|-----------|
| H  | -0.339824 | 2.463545  | -1.443617 |
| H  | -2.058364 | 2.026381  | -1.274203 |
| Br | -2.025564 | -0.513606 | 0.315140  |
| Bq | 0.272176  | -5.456981 | 5.681132  |
| Bq | 0.316086  | -5.322727 | 5.539542  |
| Bq | 0.359997  | -5.188474 | 5.397951  |
| Bq | 0.403908  | -5.054220 | 5.256361  |
| Bq | 0.447819  | -4.919966 | 5.114771  |
| Bq | 0.491729  | -4.785712 | 4.973181  |
| Bq | 0.535640  | -4.651458 | 4.831591  |
| Bq | 0.579551  | -4.517205 | 4.690001  |
| Bq | 0.623461  | -4.382951 | 4.548411  |
| Bq | 0.667372  | -4.248697 | 4.406821  |
| Bq | 0.711283  | -4.114443 | 4.265231  |
| Bq | 0.755194  | -3.980189 | 4.123640  |
| Bq | 0.799104  | -3.845935 | 3.982050  |
| Bq | 0.843015  | -3.711682 | 3.840460  |
| Bq | 0.886926  | -3.577428 | 3.698870  |
| Bq | 0.930837  | -3.443174 | 3.557280  |
| Bq | 0.974747  | -3.308920 | 3.415690  |
| Bq | 1.018658  | -3.174666 | 3.274100  |
| Bq | 1.062569  | -3.040413 | 3.132510  |
| Bq | 1.106479  | -2.906159 | 2.990920  |
| Bq | 1.150390  | -2.771905 | 2.849329  |
| Bq | 1.194301  | -2.637651 | 2.707739  |
| Bq | 1.238212  | -2.503397 | 2.566149  |
| Bq | 1.282122  | -2.369143 | 2.424559  |
| Bq | 1.326033  | -2.234890 | 2.282969  |
| Bq | 1.369944  | -2.100636 | 2.141379  |
| Bq | 1.413855  | -1.966382 | 1.999789  |
| Bq | 1.457765  | -1.832128 | 1.858199  |
| Bq | 1.501676  | -1.697874 | 1.716609  |
| Bq | 1.545587  | -1.563621 | 1.575018  |
| Bq | 1.589498  | -1.429367 | 1.433428  |
| Bq | 1.633408  | -1.295113 | 1.291838  |
| Bq | 1.677319  | -1.160859 | 1.150248  |
| Bq | 1.721230  | -1.026605 | 1.008658  |
| Bq | 1.765140  | -0.892351 | 0.867068  |

|    |          |           |           |
|----|----------|-----------|-----------|
| Bq | 1.809051 | -0.758098 | 0.725478  |
| Bq | 1.852962 | -0.623844 | 0.583888  |
| Bq | 1.896873 | -0.489590 | 0.442298  |
| Bq | 1.940783 | -0.355336 | 0.300707  |
| Bq | 1.984694 | -0.221082 | 0.159117  |
| Bq | 2.028605 | -0.086828 | 0.017527  |
| Bq | 2.072516 | 0.047425  | -0.124063 |
| Bq | 2.116426 | 0.181679  | -0.265653 |
| Bq | 2.160337 | 0.315933  | -0.407243 |
| Bq | 2.204248 | 0.450187  | -0.548833 |
| Bq | 2.248158 | 0.584441  | -0.690423 |
| Bq | 2.292069 | 0.718694  | -0.832014 |
| Bq | 2.335980 | 0.852948  | -0.973604 |
| Bq | 2.379891 | 0.987202  | -1.115194 |
| Bq | 2.423801 | 1.121456  | -1.256784 |
| Bq | 2.467712 | 1.255710  | -1.398374 |
| Bq | 2.511623 | 1.389964  | -1.539964 |
| Bq | 2.555534 | 1.524217  | -1.681554 |
| Bq | 2.599444 | 1.658471  | -1.823144 |
| Bq | 2.643355 | 1.792725  | -1.964734 |
| Bq | 2.687266 | 1.926979  | -2.106325 |
| Bq | 2.731177 | 2.061233  | -2.247915 |
| Bq | 2.775087 | 2.195486  | -2.389505 |
| Bq | 2.818998 | 2.329740  | -2.531095 |
| Bq | 2.862909 | 2.463994  | -2.672685 |
| Bq | 2.906819 | 2.598248  | -2.814275 |
| Bq | 2.950730 | 2.732502  | -2.955865 |
| Bq | 2.994641 | 2.866756  | -3.097455 |
| Bq | 3.038552 | 3.001009  | -3.239045 |
| Bq | 3.082462 | 3.135263  | -3.380636 |
| Bq | 3.126373 | 3.269517  | -3.522226 |
| Bq | 3.170284 | 3.403771  | -3.663816 |
| Bq | 3.214195 | 3.538025  | -3.805406 |
| Bq | 3.258105 | 3.672278  | -3.946996 |
| Bq | 3.302016 | 3.806532  | -4.088586 |
| Bq | 3.345927 | 3.940786  | -4.230176 |
| Bq | 3.389838 | 4.075040  | -4.371766 |
| Bq | 3.433748 | 4.209294  | -4.513356 |

|    |          |          |           |
|----|----------|----------|-----------|
| Bq | 3.477659 | 4.343548 | -4.654947 |
| Bq | 3.521570 | 4.477801 | -4.796537 |
| Bq | 3.565480 | 4.612055 | -4.938127 |
| Bq | 3.609391 | 4.746309 | -5.079717 |
| Bq | 3.653302 | 4.880563 | -5.221307 |
| Bq | 3.697213 | 5.014817 | -5.362897 |
| Bq | 3.741123 | 5.149070 | -5.504487 |
| Bq | 3.785034 | 5.283324 | -5.646077 |

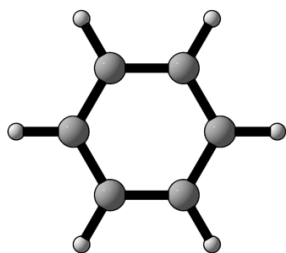
## 4 Cartesian Coordinates

Quantum chemical computations in gas phase have been performed with the Gaussian 16 (Revision B.01) suite of programs. Charge, spin multiplicity, number of imaginary frequencies, energies (in Hartree) and Cartesian coordinates (in Å) of computed geometries are listed in this section.

**Table S3.** List of calculated geometries.

| <b>Molecule</b>       | <b>Page</b> |
|-----------------------|-------------|
| Benzene               | S12         |
| Pentalene             | S12         |
| (1-Bromoethyl)benzene | S13         |
| [8]CPP                | S14         |
| [8]LPP                | S16         |
| Intricarene           | S18         |

## Benzene



#p opt freq b3lyp/6-31g (d)

Charge = 0, Multiplicity = 1, Point group =  $C_1$

Number of imaginary frequencies = 0

Electronic Energy = -232.248650991 Hartree

Sum of electronic and zero-point Energies = -232.147897 Hartree

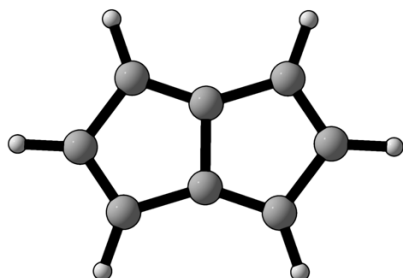
Sum of electronic and thermal Energies = -232.143508 Hartree

Sum of electronic and thermal Enthalpies = -232.142564 Hartree

Sum of electronic and thermal Free Energies = -232.175359 Hartree

| Cartesian Coordinates |           |           |           | Cartesian Coordinates |           |           |           |
|-----------------------|-----------|-----------|-----------|-----------------------|-----------|-----------|-----------|
| Atoms                 | X         | Y         | Z         | Atoms                 | X         | Y         | Z         |
| C                     | -1.339236 | -0.395853 | 0.000005  | C                     | -0.326685 | -1.357730 | 0.000069  |
| C                     | 1.012421  | -0.961878 | -0.000055 | C                     | 1.339202  | 0.395965  | 0.000008  |
| C                     | 0.326798  | 1.357700  | 0.000062  | C                     | -1.012501 | 0.961798  | -0.000058 |
| H                     | -2.381330 | -0.704010 | -0.000066 | H                     | -0.581086 | -2.414245 | 0.000079  |
| H                     | 1.800375  | -1.710234 | -0.000154 | H                     | 2.381367  | 0.703858  | 0.000007  |
| H                     | 0.580952  | 2.414267  | 0.000018  | H                     | -1.800276 | 1.710353  | -0.000063 |

## Pentalene



#p opt freq b3lyp/6-31+g (d)

Charge = 0, Multiplicity = 1, Point group =  $C_1$

Number of imaginary frequencies = 0

Electronic Energy = -308.382021547 Hartree

Sum of electronic and zero-point Energies = -308.271778 Hartree

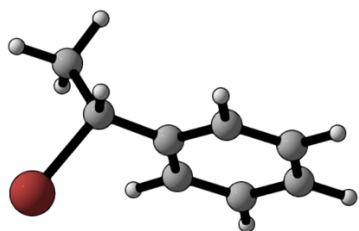
Sum of electronic and thermal Energies = -308.266098 Hartree

Sum of electronic and thermal Enthalpies = -308.265153 Hartree

Sum of electronic and thermal Free Energies = -308.301277 Hartree

| Cartesian Coordinates |           |           |           | Cartesian Coordinates |           |           |           |
|-----------------------|-----------|-----------|-----------|-----------------------|-----------|-----------|-----------|
| Atoms                 | X         | Y         | Z         | Atoms                 | X         | Y         | Z         |
| H                     | -1.541646 | 2.259359  | 0.000128  | C                     | -1.228149 | 1.219761  | 0.000028  |
| C                     | 0.037944  | 0.728591  | -0.000108 | C                     | -0.037944 | -0.728591 | -0.000106 |
| C                     | -1.465929 | -1.099261 | 0.000043  | C                     | -2.174368 | 0.060138  | 0.000014  |
| H                     | -1.868928 | -2.105044 | 0.000047  | H                     | -3.254159 | 0.157549  | 0.000035  |
| C                     | 1.228149  | -1.219761 | -0.000030 | C                     | 2.174368  | -0.060138 | 0.000105  |
| C                     | 1.465929  | 1.099261  | -0.000022 | H                     | 1.541646  | -2.259359 | 0.000069  |
| H                     | 3.254159  | -0.157549 | 0.000215  | H                     | 1.868928  | 2.105044  | -0.000033 |

### (1-Bromoethyl)benzene



#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group =  $C_1$

Number of imaginary frequencies = 0

Electronic Energy = -2881.9871006 Hartree

Sum of electronic and zero-point Energies = -2881.839424 Hartree

Sum of electronic and thermal Energies = -2881.830929 Hartree

Sum of electronic and thermal Enthalpies = -2881.829984 Hartree

Sum of electronic and thermal Free Energies = -2881.874497 Hartree

| Cartesian Coordinates |           |           |           | Cartesian Coordinates |           |           |           |
|-----------------------|-----------|-----------|-----------|-----------------------|-----------|-----------|-----------|
| Atoms                 | <i>X</i>  | <i>Y</i>  | <i>Z</i>  | Atoms                 | <i>X</i>  | <i>Y</i>  | <i>Z</i>  |
| H                     | -1.336686 | 1.593573  | -1.700799 | C                     | -1.644023 | 0.860759  | -0.958039 |
| C                     | -2.436153 | -1.023041 | 0.934252  | C                     | -0.709486 | 0.380805  | -0.033205 |
| C                     | -2.965898 | 0.410664  | -0.934686 | C                     | -3.365441 | -0.531002 | 0.012637  |
| C                     | -1.119193 | -0.571753 | 0.911178  | H                     | -3.678811 | 0.793674  | -1.659899 |
| H                     | -4.392198 | -0.886242 | 0.030798  | H                     | -0.392663 | -0.977428 | 1.609798  |
| H                     | -2.738566 | -1.764968 | 1.668450  | C                     | 0.688271  | 0.939771  | -0.033079 |
| H                     | 0.845039  | 1.561075  | -0.915349 | C                     | 1.067929  | 1.699453  | 1.233267  |
| H                     | 0.980887  | 1.064089  | 2.119229  | H                     | 0.388795  | 2.552569  | 1.357761  |
| H                     | 2.094056  | 2.072410  | 1.173348  | Br                    | 2.032403  | -0.542934 | -0.299351 |

[8]CPP



#p opt freq rwb97xd/6-31g(d)

Charge = 0, Multiplicity = 1, Point group =  $C_1$

Number of imaginary frequencies = 0

Electronic Energy = -1847.71625488 Hartree

Sum of electronic and zero-point Energies = -1847.061537 Hartree

Sum of electronic and thermal Energies = -1847.027355 Hartree

Sum of electronic and thermal Enthalpies = -1847.026410 Hartree

Sum of electronic and thermal Free Energies = -1847.126415 Hartree

| Atoms | Cartesian Coordinates |           |           | Atoms | Cartesian Coordinates |           |           |
|-------|-----------------------|-----------|-----------|-------|-----------------------|-----------|-----------|
|       | X                     | Y         | Z         |       | X                     | Y         | Z         |
| H     | 3.456122              | 5.330052  | 1.789288  | C     | 3.633979              | 4.680153  | 0.936594  |
| C     | 4.125308              | 3.008015  | -1.211276 | C     | 2.793199              | 4.756153  | -0.182810 |
| C     | 4.621335              | 3.706006  | 1.022804  | C     | 4.808999              | 2.779917  | -0.011884 |
| C     | 3.133626              | 3.976005  | -1.293871 | H     | 5.192054              | 3.607298  | 1.942568  |
| H     | 2.514056              | 4.011866  | -2.184962 | H     | 4.253500              | 2.316471  | -2.038984 |
| C     | 1.415569              | 5.314617  | -0.109158 | C     | -1.417681             | 5.314052  | -0.109199 |
| C     | 0.692327              | 5.113129  | 1.072060  | C     | 0.694220              | 5.723635  | -1.239222 |
| C     | -0.696459             | 5.723353  | -1.239241 | C     | -0.694400             | 5.112870  | 1.072043  |
| H     | 1.211464              | 4.799163  | 1.972390  | H     | 1.222791              | 5.966602  | -2.157290 |
| H     | -1.225101             | 5.966103  | -2.157325 | H     | -1.213438             | 4.798702  | 1.972361  |
| C     | -2.795079             | 4.755033  | -0.182931 | C     | -4.810120             | 2.778008  | -0.012093 |
| C     | -3.636002             | 4.678827  | 0.936354  | C     | -3.135000             | 3.974598  | -1.293943 |
| C     | -4.126298             | 3.006210  | -1.211391 | C     | -4.622985             | 3.704299  | 1.022517  |
| H     | -3.458543             | 5.328904  | 1.788994  | H     | -2.515294             | 4.010616  | -2.184935 |
| H     | -4.254069             | 2.314503  | -2.039029 | H     | -5.193837             | 3.605489  | 1.942185  |
| C     | -5.364336             | 1.417231  | 0.220409  | C     | -5.357015             | -1.417454 | 0.351923  |
| C     | -5.979522             | 0.648476  | -0.775450 | C     | -4.945501             | 0.747832  | 1.375226  |
| C     | -4.935923             | -0.636714 | 1.436088  | C     | -5.986043             | -0.740654 | -0.704434 |
| H     | -6.385943             | 1.135921  | -1.657923 | H     | -4.471861             | 1.308396  | 2.175636  |
| H     | -4.451019             | -1.102144 | 2.287070  | H     | -6.412708             | -1.300407 | -1.531853 |
| H     | 4.450823              | -1.100667 | 2.286717  | C     | 4.935786              | -0.634907 | 1.435941  |
| C     | 5.979500              | 0.650927  | -0.775165 | C     | 5.357496              | -1.415331 | 0.351822  |
| C     | 4.944831              | 0.749638  | 1.375228  | C     | 5.363684              | 1.419305  | 0.220571  |
| C     | 5.986586              | -0.738210 | -0.704272 | H     | 4.470820              | 1.309934  | 2.175607  |
| H     | 6.413829              | -1.297733 | -1.531547 | H     | 6.385968              | 1.138631  | -1.657471 |
| C     | 4.823200              | -2.801155 | 0.218017  | C     | 2.814202              | -4.769824 | -0.150112 |
| H     | -4.978348             | -2.872037 | -1.935907 | H     | -4.530422             | -3.199138 | 2.323680  |

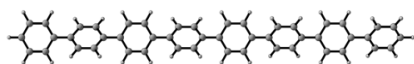
| Cartesian Coordinates |           |           |           | Cartesian Coordinates |           |           |           |
|-----------------------|-----------|-----------|-----------|-----------------------|-----------|-----------|-----------|
| Atoms                 | <i>X</i>  | <i>Y</i>  | <i>Z</i>  | Atoms                 | <i>X</i>  | <i>Y</i>  | <i>Z</i>  |
| C                     | 4.538146  | -3.320138 | -1.052544 | C                     | 4.270812  | -3.485421 | 1.309785  |
| C                     | 3.271075  | -4.432762 | 1.130118  | C                     | 3.570811  | -4.298629 | -1.231094 |
| H                     | 4.979379  | -2.869896 | -1.936073 | H                     | 4.531856  | -3.197401 | 2.323507  |
| H                     | 2.753057  | -4.811999 | 2.006199  | H                     | 3.323250  | -4.608433 | -2.242256 |
| C                     | 1.422382  | -5.262116 | -0.339182 | C                     | -1.420241 | -5.262653 | -0.339061 |
| C                     | 0.697394  | -5.959669 | 0.637505  | C                     | 0.692737  | -4.734742 | -1.411310 |
| C                     | -0.690887 | -4.735022 | -1.411259 | C                     | -0.694919 | -5.959932 | 0.637565  |
| H                     | 1.221707  | -6.450605 | 1.453152  | H                     | 1.202604  | -4.179628 | -2.191237 |
| H                     | -1.201060 | -4.180157 | -2.191166 | H                     | -1.218981 | -6.451073 | 1.453251  |
| C                     | -2.812255 | -4.770911 | -0.149956 | C                     | -4.822079 | -2.803053 | 0.218184  |
| C                     | -3.269201 | -4.433973 | 1.130277  | C                     | -3.569105 | -4.300072 | -1.230915 |
| C                     | -4.536855 | -3.321991 | -1.052360 | C                     | -4.269330 | -3.487046 | 1.309947  |
| H                     | -2.751008 | -4.812971 | 2.006360  | H                     | -3.321438 | -4.609788 | -2.242080 |

# [8]LPP

#p opt freq wb97xd/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C<sub>1</sub>

Number of imaginary frequencies = 0



Electronic Energy = -1849.02620993 Hartree

Sum of electronic and zero-point Energies = -1848.349904 Hartree

Sum of electronic and thermal Energies = -1848.313242 Hartree

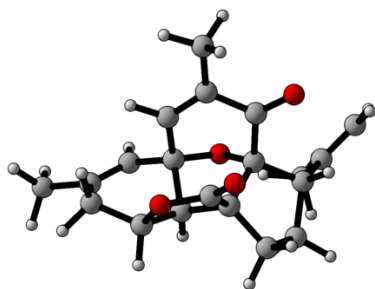
Sum of electronic and thermal Enthalpies = -1848.312298 Hartree

Sum of electronic and thermal Free Energies = -1848.424957 Hartree

| Cartesian Coordinates |            |           |           | Cartesian Coordinates |            |           |           |
|-----------------------|------------|-----------|-----------|-----------------------|------------|-----------|-----------|
| Atoms                 | X          | Y         | Z         | Atoms                 | X          | Y         | Z         |
| H                     | 5.240133   | -1.614293 | -1.412016 | C                     | 5.774598   | -0.914553 | -0.775484 |
| C                     | 7.162875   | 0.914240  | 0.775720  | C                     | 5.054027   | -0.000002 | -0.000003 |
| C                     | 7.162876   | -0.914244 | -0.775724 | C                     | 7.883627   | -0.000003 | -0.000001 |
| C                     | 5.774597   | 0.914550  | 0.775477  | H                     | 7.697262   | -1.613672 | -1.412641 |
| H                     | 5.240131   | 1.614290  | 1.412007  | H                     | 7.697260   | 1.613668  | 1.412637  |
| C                     | 3.570906   | -0.000001 | -0.000004 | C                     | 0.741666   | 0.000000  | -0.000005 |
| C                     | 2.850422   | -1.199056 | 0.002539  | C                     | 2.850424   | 1.199054  | -0.002547 |
| C                     | 1.462135   | 1.199060  | -0.001361 | C                     | 1.462134   | -1.199061 | 0.001353  |
| H                     | 3.385229   | -2.144630 | 0.020719  | H                     | 3.385231   | 2.144628  | -0.020725 |
| H                     | 0.927342   | 2.144700  | 0.013752  | H                     | 0.927339   | -2.144699 | -0.013759 |
| C                     | -0.741444  | 0.000001  | -0.000005 | C                     | -3.571027  | 0.000001  | -0.000004 |
| C                     | -1.462030  | -0.912115 | 0.778336  | C                     | -1.462030  | 0.912116  | -0.778346 |
| C                     | -2.850303  | 0.911842  | -0.778565 | C                     | -2.850303  | -0.911841 | 0.778556  |
| H                     | -0.927585  | -1.609854 | 1.417067  | H                     | -0.927585  | 1.609855  | -1.417077 |
| H                     | -3.384635  | 1.609288  | -1.417698 | H                     | -3.384635  | -1.609287 | 1.417689  |
| C                     | -5.054065  | 0.000000  | -0.000004 | C                     | -7.883667  | 0.000000  | -0.000002 |
| C                     | -5.774737  | -1.198959 | 0.014487  | C                     | -5.774737  | 1.198960  | -0.014495 |
| C                     | -7.163005  | 1.198949  | -0.014589 | C                     | -7.163005  | -1.198949 | 0.014582  |
| H                     | -5.240333  | -2.144678 | -0.007175 | H                     | -5.240333  | 2.144678  | 0.007167  |
| H                     | -7.697580  | 2.144575  | 0.006796  | H                     | -7.697580  | -2.144574 | -0.006803 |
| C                     | -9.366729  | 0.000001  | -0.000002 | C                     | -12.196232 | 0.000003  | 0.000002  |
| C                     | -10.087306 | -0.909075 | 0.781815  | C                     | -10.087306 | 0.909078  | -0.781814 |
| C                     | -11.475656 | 0.909038  | -0.781730 | C                     | -11.475654 | -0.909034 | 0.781735  |
| H                     | -9.552809  | -1.604369 | 1.423193  | H                     | -9.552812  | 1.604373  | -1.423193 |
| H                     | -12.010396 | 1.604448  | -1.422747 | H                     | -12.010395 | -1.604443 | 1.422752  |
| C                     | 9.366710   | -0.000002 | -0.000001 | C                     | 12.196235  | -0.000001 | 0.000002  |
| C                     | 10.087343  | 1.198984  | 0.010749  | C                     | 10.087344  | -1.198987 | -0.010750 |

| Cartesian Coordinates |            |           |           | Cartesian Coordinates |            |           |           |
|-----------------------|------------|-----------|-----------|-----------------------|------------|-----------|-----------|
| Atoms                 | <i>X</i>   | <i>Y</i>  | <i>Z</i>  | Atoms                 | <i>X</i>   | <i>Y</i>  | <i>Z</i>  |
| C                     | 11.475695  | -1.198912 | -0.010988 | C                     | 11.475693  | 1.198910  | 0.010990  |
| H                     | 9.552932   | 2.144642  | -0.013785 | H                     | 9.552934   | -2.144646 | 0.013785  |
| H                     | 12.010520  | -2.144344 | 0.012818  | H                     | 12.010519  | 2.144342  | -0.012811 |
| C                     | 13.680370  | 0.000001  | 0.000005  | C                     | 16.487684  | 0.000004  | 0.000009  |
| C                     | 14.396373  | 0.903527  | 0.793218  | C                     | 14.396378  | -0.903524 | -0.793207 |
| C                     | 15.786965  | -0.904570 | -0.792966 | C                     | 15.786960  | 0.904576  | 0.792981  |
| H                     | 13.856485  | 1.595773  | 1.433492  | H                     | 13.856493  | -1.595770 | -1.433484 |
| H                     | 16.324690  | -1.608937 | -1.420934 | H                     | 16.324682  | 1.608945  | 1.420951  |
| H                     | 17.573563  | 0.000005  | 0.000010  | C                     | -13.680384 | 0.000003  | 0.000004  |
| C                     | -16.487682 | -0.000002 | 0.000008  | C                     | -14.396388 | -1.202329 | 0.000338  |
| C                     | -14.396391 | 1.202328  | -0.000327 | C                     | -15.786980 | 1.202960  | -0.001249 |
| C                     | -15.786974 | -1.202965 | 0.001264  | H                     | -13.856520 | -2.144972 | -0.023883 |
| H                     | -13.856530 | 2.144975  | 0.023892  | H                     | -16.324742 | 2.146563  | 0.005642  |
| H                     | -16.324737 | -2.146568 | -0.005626 | H                     | -17.573562 | -0.000007 | 0.000010  |

## Intricarene



#p opt rb3lyp/6-31+g (d, p)

Charge = 0, Multiplicity = 1, Point group =  $C_1$

Electronic Energy = -1076.27942875 Hartree

| Cartesian Coordinates |           |           |           | Cartesian Coordinates |           |           |           |
|-----------------------|-----------|-----------|-----------|-----------------------|-----------|-----------|-----------|
| Atoms                 | X         | Y         | Z         | Atoms                 | X         | Y         | Z         |
| C                     | -0.928494 | -0.608253 | -0.662992 | C                     | -2.280182 | 0.041908  | 1.378973  |
| C                     | -3.221474 | -1.366511 | -0.600570 | C                     | -3.475197 | -0.254184 | 0.428266  |
| C                     | -1.997769 | -1.554766 | -1.131478 | C                     | -1.012646 | -0.608221 | 0.891509  |
| H                     | -2.532819 | -0.258527 | 2.401108  | H                     | -3.725362 | 0.679138  | -0.093799 |
| H                     | -4.361805 | -0.507950 | 1.018359  | H                     | -1.789178 | -2.308756 | -1.885335 |
| C                     | -4.421757 | -2.180565 | -1.006012 | H                     | -4.850029 | -2.706904 | -0.143133 |
| H                     | -4.165684 | -2.921788 | -1.768008 | H                     | -5.216343 | -1.539013 | -1.409177 |
| O                     | -1.927183 | 1.454806  | 1.437729  | C                     | -0.559842 | 1.629437  | 1.444299  |
| O                     | -0.064559 | 2.714122  | 1.614657  | C                     | -1.101038 | 0.737253  | -1.375343 |
| C                     | -0.074296 | 1.580450  | -1.599429 | C                     | 1.212725  | 1.277390  | -0.914789 |
| O                     | 2.208696  | 1.973991  | -1.017859 | O                     | 0.451844  | -0.978985 | -0.790962 |
| C                     | 1.129387  | 0.047811  | -0.005546 | H                     | -0.914775 | -1.647665 | 1.208583  |
| H                     | -2.088587 | 0.970560  | -1.763002 | C                     | -0.163714 | 2.843569  | -2.404764 |
| H                     | 0.555862  | 2.823692  | -3.230440 | H                     | 0.091886  | 3.710256  | -1.786058 |
| H                     | -1.167804 | 2.980391  | -2.814852 | C                     | 0.155656  | 0.277615  | 1.263756  |
| C                     | 2.467331  | -0.447080 | 0.573650  | H                     | 3.027418  | 0.473600  | 0.774857  |
| C                     | 1.018413  | -0.105841 | 2.486285  | H                     | 0.416539  | -0.538513 | 3.293309  |
| H                     | 1.495761  | 0.799822  | 2.876804  | C                     | 2.077574  | -1.069445 | 1.931968  |
| H                     | 2.939017  | -1.175520 | 2.598699  | H                     | 1.646712  | -2.067044 | 1.790092  |
| C                     | 3.322357  | -1.304965 | -0.350746 | C                     | 4.348202  | -0.752824 | -1.013406 |
| H                     | 4.971604  | -1.343452 | -1.679735 | H                     | 4.580816  | 0.304141  | -0.925823 |
| C                     | 3.001321  | -2.775919 | -0.480241 | H                     | 3.652010  | -3.248398 | -1.221454 |
| H                     | 1.960255  | -2.927863 | -0.781698 | H                     | 3.144779  | -3.302835 | 0.471838  |